

1. A server sends 1024 Mb of data to a client over a 16.4 Mbps link with 2.5% of packet loss. The server sends the data in packets of 8 Mb and after sending a packet awaits to receive an acknowledgment packet of 8 bytes from the client before sending the next packet. If it takes 10 minutes to complete the transfer of data, determine the latency of the link.

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(Packets to send) Packet loss percentage

$$PTS = \frac{1024Mb}{8Mb} + \left(\frac{1024Mb}{8Mb} * 0.025 \right) = 131.2 \approx 132$$

(Time to send a packet)

$$TTP = \frac{8Mb}{16.4 Mb/s} = 0.4878s$$

ACK = 8B = 64b

(Time to send acknowledgment)

$$TTSACK = \frac{64b}{16.4 \times 10^6 b/s} = 3.9 \times 10^{-6}s \text{ (This time is negligible because it's small)}$$

(Total latency)

$$Tlatency = 10m = 600s$$

(Latency of the link)

$$600s = 132(3.9 \times 10^{-6}s + 0.4878s + Llink)$$

$$\frac{600s}{132} = 0.4878039s + Llink$$

$$4.545454s = 0.4878039s + Llink$$

$$4.545454s - 0.4878039s = Llink$$

$$4.0576501s = Llink$$

$$\approx 4.06s = Llink$$

The latency of the link is approximately 4.06s.

2. Explain the logic behind the phrase:

“You can buy more bandwidth, but you cannot buy less delay.” Exemplify and motivate your answer.

La frase quiere decir que aunque se compre más ancho de banda no se va a poder reducir el delay, esto cobra sentido ya que aumentar el ancho de banda es más fácil que reducir el delay, por ejemplo: si se va en un tren y cada vagón puede llevar 10 pasajeros y el tiempo de viaje de ese punto a otro es de 10 minutos, si se quisiera llevar más pasajeros simplemente se podrían hacer vagones que lleven a 20 personas o poner mas vagones, sin embargo el tiempo de viaje siempre será 10 minutos ya que es lo que se tarda en trasladarse, reducir ese tiempo de viaje puede ser complejo. Así mismo es el ancho de banda y el delay, se puede tener un ancho de banda muy grande pero el delay siempre va a ser el mismo.

3. Compare Datagram to the Virtual Circuit networks with respect to:
– circuit setup; addressing scheme; routing; router failure; Quality of Service;

Aspect	Datagram	Virtual Circuit Networks
Circuit setup	Each packet is switched independently	All the packets are transmitted along the pre-established path. This occurs before the transmission phase.
Addressing Scheme	Each Packet header contains a complete destination address	Packet headers don't need to contain the full destination address of the packet - only contain virtual circuit identifier (VCI)
routing	Router looks at the packet's destination address and searches its current routing table to determine the next hop (for each packet)	The route is determined during circuit establishment and remains fixed during the session.
Router failure	Routes may change during session (greater resilience)	The connection is interrupted if a router along the route fails; it requires reestablishing the circuit.
Quality of Service	Does not guarantee order, minimum latency or packet loss	It offers guaranteed in-order delivery, low loss, and controlled latency (ideal for time-sensitive applications).